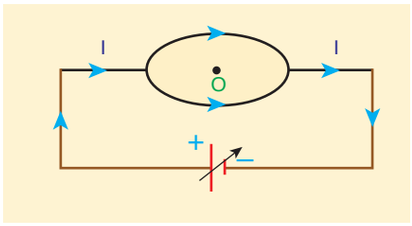
$$\vec{B} = \frac{\mu_\circ I}{2\pi} \frac{R^2}{\left(R^2 + Z^2\right)^{\frac{3}{2}}} \hat{k} \qquad (3.40)$$

Note that the magnetic field $\vec{B}$ points along the direction from the point O to P. Suppose if the current flows in clockwise direction, then magnetic field points in the direction from the point P to O.

EXAMPLE 3.17

What is the magnetic field at the center of the loop shown in figure?



Solution

The magnetic field due to current in the upper hemisphere and lower hemisphere of the circular coil are equal in magnitude but opposite in direction. Hence, the net magnetic field at the center of the loop (at point O) is zero $\vec{B} = \vec{0}$.

3.8.4 Current loop as a magnetic dipole

The magnetic field from the centre of a circular loop of radius R along the axis is given by

$$\vec{B} = \frac{\mu_\circ I}{2} \frac{R^2}{\left(R^2 + z^2\right)^{\frac{3}{2}}} \hat{k}$$

At larger distance Z >> R, therefore $R^2 + Z^2 \approx Z^2$, we have

$$\vec{B} = \frac{\mu_\circ I}{2} \frac{R^2}{Z^3} \hat{k} \qquad (3.41)$$

Let A be the area of the circular loop $A = \pi R^2$. So rewriting the equation (3.41) in terms of area of the loop, we have

$$\vec{B} = \frac{\mu_\circ I}{2\pi} \frac{A}{Z^3} \hat{k}$$

(multiply and divide by 2)

$$\vec{B} = \frac{\mu_\circ}{4\pi} \frac{2IA}{Z^3} \hat{k} \qquad (3.42)$$

Comparing equation (3.42) with equation (3.14) dimensionally, we get

$$p_m = I\,A$$

where p_m is called magnetic dipole moment. In vector notation,

$$\vec{p}_m = I\,\vec{A} \qquad (3.43)$$

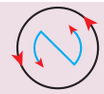
This implies that a current carrying circular loop behaves as a magnetic dipole of magnetic moment $\vec{p}_m$. So, **the magnetic dipole moment of any current loop is equal to the product of the current and area of the loop.**

Right hand thumb rule

In order to determine the direction of magnetic moment, we use right hand thumb rule (mnemonic) which states that

If we curl the fingers of right hand in the direction of current in the loop, then the stretched thumb gives the direction of the magnetic moment associated with the loop.

Table 3.3 End rule – polarity with direction of current in circular loop

Current in circular loop	Polarity	Picture
Anti-clockwise current	North Pole	 Anti-clockwise current Polarity: North Pole
Clockwise current	South Pole	 Clockwise current Polarity: South Pole

UNIT-3(XII-Physics_Vol-1).indd 167 04-03-2019 10:34:33